



# WORK BASED LEARNING LOGBOOK

|                         |                                   |
|-------------------------|-----------------------------------|
| <b>Semester/Session</b> | <b>Semester 7 / 2024/2025</b>     |
| <b>Student Name</b>     | <b>Siti Rabaah binti Mohd Zin</b> |
| <b>Matrix No.</b>       | <b>AW210003</b>                   |
| <b>Faculty Name</b>     | <b>FAST</b>                       |
| <b>Company Name</b>     | <b>Onelab Testing Sdn Bhd</b>     |



**JABATAN FIZIK DAN KIMIA  
FAKULTI SAINS GUNAAN DAN TEKNOLOGI  
PROGRAM SARJANA MUDA KIMIA INDUSTRI DENGAN KEPUJIAN**

**A. Student's particulars (To be filled by student)**

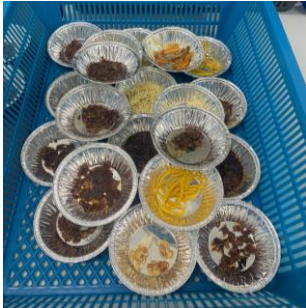
Student's name : Siti Rabaah binti Mohd Zin  
Matric no. : AW210003  
Semester/session : 7/2024/2025  
Company's name & Address : Onelab Testing Sdn. Bhd., No 10A, Jalan Tanjong 2,  
Taman Desa Cemerlang, 81800 Ulu Tiram, Johor

**B. Instruction: Students must complete this report weekly. This report must be endorsed by the industrial supervisor and uploaded to the Author system.**

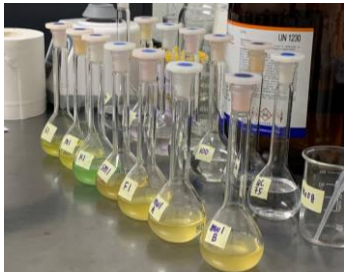
Date/week:  
**WEEK 8 (4-8/11/2024)**

| 4 November 2024                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| <b>BWK40103</b><br>(Quality Control and Assurance)        | <ul style="list-style-type: none"><li>For every batch of 10 samples, create and analyze a duplicate of one randomly selected sample within that batch to confirm the method's reliability and accuracy.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>BWK40204</b><br>(Industrial Safety)                    | <ul style="list-style-type: none"><li>Wear a lab coat when working with chemicals to protect skin and clothing from spills and splashes, and gloves to protect from direct contact with hazardous substances.</li><li>Wear heat-resistant gloves to avoid burns when placing the flask in or removing it from the water bath.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>BWK40304</b><br>(Industrial Technology and Operations) | <p><b>Determining Solubility of Sudan Dyes</b></p> <ul style="list-style-type: none"><li>Sudan dyes and Para Red are dissolved in a mobile phase mixture with a 60:40 ratio of acetonitrile to methanol.</li><li>Adding water to Sudan dyes and sonication for 15 minutes does not achieve complete dissolution. Sudan dyes are hydrophobic, poorly soluble in water due to their non-polar chemical structure. Water (polar solvent) is not effective in dissolving non-polar compounds like Sudan dyes. When water is added, it does not interact well with the hydrophobic regions of these dyes, leading to incomplete dissolution. Even after sonication, which agitates the solution and can sometimes aid in dissolving substances, the dyes remain undissolved because the polarity mismatch between the dyes and water remains unresolved.</li><li>Therefore, a more non-polar or mixed organic solvent is needed to dissolve Sudan dyes completely.</li><li>Sudan III and Sudan IV exhibit overlapping peaks at identical</li></ul> |

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|                                                            | <p>retention times.</p> <p><b>Sample preparation for caffeine analysis</b></p> <ul style="list-style-type: none"> <li>• Prepare a stock solution of caffeine and dilute it to create five calibration standards.</li> <li>• Weigh the sample, magnesium oxide and add hot water, then swirl to mix. Magnesium oxide (MgO) is used in caffeine analysis to extract caffeine from coffee samples and to help maintain the analytical column's lifetime.</li> <li>• Place the solution in a water bath at a controlled temperature for 1 hour to facilitates the release of caffeine from the sample matrix, aiding its extraction.</li> <li>• Swirling every 15 minutes, then allow it to cool to room temperature. Add another volume of water to the flask, filter the solution, and transfer it to a vial.</li> </ul>                                                      |
| <b>5 November 2024</b>                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <p><b>BWK40103</b><br/>(Quality Control and Assurance)</p> | <ul style="list-style-type: none"> <li>• For every batch of 10 samples, create and analyze a duplicate of one randomly selected sample within that batch to confirm the method's reliability and accuracy.</li> <li>• Assess peak shapes and resolution between Sudan III and Sudan IV in each chromatographic run to monitor for any drift or changes in peak overlap.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <p><b>BWK40204</b><br/>(Industrial Safety)</p>             | <p><b>Safety Training Module II</b></p> <ul style="list-style-type: none"> <li>• A safety training module focused on hazard identification, risk assessment, and risk control is designed to equip employees and lab personnel with essential skills for recognizing and managing workplace hazards.</li> <li>• Hazard identification: Identify and categorize potential hazards (<i>chemical, physical, biological, ergonomic, psychosocial</i>) and sources of potential harm (<i>equipment, man, machine, material and method</i>).</li> <li>• Risk assessment: Understanding risk assessment principles and frameworks such as likelihood and severity matrix to determine which risks need immediate attention based on their severity and likelihood.</li> <li>• Risk control: Hierarchy of controls for managing hazards: elimination, substitution, engineering controls, administrative controls, and personal protective equipment (PPE).</li> </ul> |

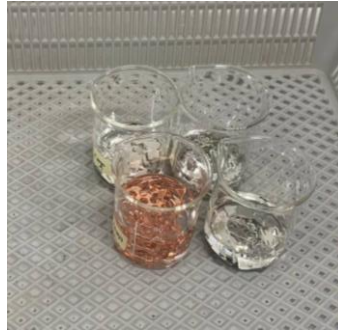
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| <b>BWK40304</b><br>(Industrial Technology and Operations) | <p><b>Solubility Sudan Dyes</b></p> <ul style="list-style-type: none"> <li>• The solubility of Sudan III, Sudan IV, and Para Red was tested by dissolving the dyes in a mobile phase consisting of 60% methanol and 40% acetonitrile, with the addition of acetone.</li> <li>• The addition of acetone improved the solubility of the dyes compared to the mobile phase mixture alone.</li> <li>• Despite the improved solubility, Sudan III and Sudan IV continued to show overlapping peaks in chromatographic analysis.</li> </ul> <p><b>Moisture testing</b></p> <ul style="list-style-type: none"> <li>• Weigh food samples for moisture testing and dry the samples in oven for few hours to ensure that all free moisture has been fully removed, leading to a stable, accurate, and reproducible measurement.</li> <li>• Then, cooling the samples to room temperature and the final weight of moisture is obtained by constant weight techniques.</li> </ul>  |
| <b>6 November 2024</b>                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>BWK40103</b><br>(Quality Control and Assurance)        | <ul style="list-style-type: none"> <li>• Calibrate the pH meter before each testing session by using standardized buffer solutions (pH 4, 7, and 10 buffers) to ensure accuracy.</li> <li>• For every batch of 10 samples, create and analyze a duplicate of one randomly selected sample within that batch to confirm the method's reliability and accuracy.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>BWK40204</b><br>(Industrial Safety)                    | <ul style="list-style-type: none"> <li>• Wear a lab coat when working with chemicals to protect skin and clothing from spills and splashes, and gloves to protect from direct contact with hazardous substances.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>BWK40304</b><br>(Industrial Technology and Operations) | <p><b>Solubility Dyes</b></p> <ul style="list-style-type: none"> <li>• To determine the solubility of each dye, they were dissolved in a solvent mixture of methanol (60%) and acetonitrile (40%). While this solvent system was generally effective, Sudan III, Sudan IV, and Para Red did not fully dissolve under these conditions.</li> <li>• To improve solubility, a few drops of acetone were added to each solution. This adjustment enhanced dissolution but was insufficient to fully dissolve the dyes.</li> <li>• This is because of the high mass of the dye samples relative to the solvent volume presented a challenge, limiting the</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                           |

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|  | <p>complete solubility of Sudan III, Sudan IV, and Para Red in the mixture.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|  | <p><b>Sodium as sodium chloride method</b></p> <ul style="list-style-type: none"> <li>• Weigh the food sample and dissolve it in hot distilled water to ensure that the analytes (chloride ions) are fully extracted from the food matrix into the solution, making them accessible for titration.</li> <li>• Stir the solution thoroughly using a glass rod, then filter it into a separate conical flask using a filter funnel and filter paper.</li> <li>• Take the filtrate for titration against <math>\text{AgNO}_3</math> by using <math>\text{KCr}_2\text{O}_7</math> as the indicator.</li> <li>• During titration, silver ions (<math>\text{Ag}^+</math>) react with chloride ions (<math>\text{Cl}^-</math>) to form a white precipitate of silver chloride (<math>\text{AgCl}</math>). When all chloride ions have reacted, excess silver ions react with potassium dichromate, producing a red-brown color that signals the endpoint.</li> </ul> |
|  | <p><b>Water soluble matter</b></p> <ul style="list-style-type: none"> <li>• Weigh the sample and dissolve it in water. Reflux the solution for one hour, allowing it to reach a boil.</li> <li>• Filter the solution into a volumetric flask, rinsing the funnel with water until it runs clear. Top up the volumetric flask with water to the calibration mark.</li> <li>• Pipette the prepared solution into an empty beaker and heat it until only the residue remains.</li> <li>• Place the beaker in an oven to dry the residue thoroughly, then record the final weight.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                     |
|  | <p><b>Sample preparation for sodium dehydroacetate analysis</b></p> <ul style="list-style-type: none"> <li>• Weigh the sample into a centrifuge tube and add an appropriate volume of alcohol. The mixture is then subjected to sonication to facilitate extraction.</li> <li>• Following sonication, the sample is centrifuged to extract the sodium dehydroacetate from the sample matrix.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|  | <p><b>pH testing</b></p> <ul style="list-style-type: none"> <li>• Weigh food samples and dissolve them in distilled water.</li> <li>• Calibrate the pH meter using standard buffer solutions begin with the pH 4 buffer solution. Rinse the electrode with distilled water and immerse it in the pH 4 buffer for calibration purpose.</li> <li>• Rinse the electrode again and proceed with the calibration of pH 7 buffer solution.</li> <li>• Finally, rinse the electrode and calibrate with the pH 10 buffer</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                   |

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|                                                           | <p>solution.</p> <ul style="list-style-type: none"> <li>• Rinse the pH electrode with distilled water and immerse it into the prepared food sample solution.</li> <li>• Allow the reading to stabilize, then record the pH and the temperature of the solution.</li> </ul>                                                                                                                                                                                                                                     |
| <b>7 November 2024</b>                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>BWK40103</b><br>(Quality Control and Assurance)        | <ul style="list-style-type: none"> <li>• Ensure the workspace is organized, and all equipment is properly stored after use.</li> <li>• Verify the sample lab IDs in the system to check for any updates related to the food testing.</li> </ul>                                                                                                                                                                                                                                                                |
| <b>BWK40204</b><br>(Industrial Safety)                    | <ul style="list-style-type: none"> <li>• Perform the procedure in a fume hood to minimize inhalation of fumes from hydrochloric acid during the preparation of fat testing.</li> <li>• Regularly check the temperature settings of the water bath to maintain its temperature, as overheating can lead to hazardous conditions or degradation of samples</li> </ul>                                                                                                                                            |
| <b>BWK40304</b><br>(Industrial Technology and Operations) | <p><b>Sample preparation for sodium dehydroacetate</b></p> <ul style="list-style-type: none"> <li>• Weigh the sample into a centrifuge tube and add an appropriate volume of alcohol. The mixture is then subjected to sonication to facilitate extraction.</li> <li>• Following sonication, the sample is centrifuged to extract the sodium dehydroacetate from the sample matrix.</li> </ul>                             |
|                                                           | <p><b>Fat extraction</b></p> <ul style="list-style-type: none"> <li>• The total fat for food products is done via acid hydrolysis method. The food samples are added with concentrated acid and heated to certain temperature.</li> <li>• The fat will be hydrolysed, and the extraction is performed using petroleum ether.</li> <li>• After extraction for three times, the solvent is evaporated off in heated water bath and the final weight of fat is obtained by constant weight techniques.</li> </ul> |
|                                                           | <p><b>Moisture testing</b></p> <ul style="list-style-type: none"> <li>• Weigh food samples for moisture testing and dry the samples in oven for few hours to ensure that all free moisture has been fully removed, leading to a stable, accurate, and reproducible measurement.</li> <li>• Then, cool the samples to room temperature and final weight of moisture is obtained by constant weight techniques.</li> </ul>                                                                                       |
|                                                           | <b>Oil content</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

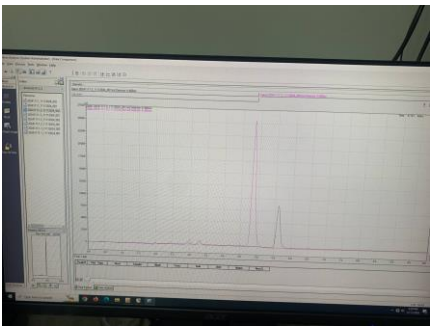
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|                                                           | <ul style="list-style-type: none"> <li>• Total oil content is done via immersing samples (usually metals) in a non-polar solvent for a few hours allows extraction of non-polar compounds. Non-polar solvents are used since they are effective at dissolving hydrophobic substances like oil.</li> <li>• Then transfer the solution to a clean beaker and remove any remaining metal ensures that no residual contaminants or unwanted particles affect the analysis. The solution then evaporated in heated water bath at certain temperature and the remaining extracted oil is placed in an oven at a controlled temperature and allow cool to room temperature.</li> <li>• Once the samples have cooled, weigh each sample and record the weight data accurately.</li> <li>• Assess the recorded weights to determine if the samples have achieved a constant weight.</li> </ul>                                                                  |
| <b>8 November 2024</b>                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>BWK40103</b><br>(Quality Control and Assurance)        | <ul style="list-style-type: none"> <li>• Perform duplicate extractions on sample to assess repeatability.</li> <li>• Monitor the endpoint of titration carefully to avoid systematic errors.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>BWK40204</b><br>(Industrial Safety)                    | <ul style="list-style-type: none"> <li>• Wear a lab coat to protect against spills and chemical exposure.</li> <li>• Always wear appropriate gloves when handling organic acids to prevent skin contact</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>BWK40304</b><br>(Industrial Technology and Operations) | <b>Sodium as sodium chloride method</b> <ul style="list-style-type: none"> <li>• Weigh the food sample and dissolve it in hot distilled water to ensure that the analytes (chloride ions) are fully extracted from the food matrix into the solution, making them accessible for titration.</li> <li>• Stir the solution thoroughly using a glass rod, then filter it into a separate conical flask using a filter funnel and filter paper.</li> <li>• Take the filtrate for titration against <math>\text{AgNO}_3</math> by using <math>\text{KCr}_2\text{O}_7</math> as the indicator.</li> <li>• During titration, silver ions (<math>\text{Ag}^+</math>) react with chloride ions (<math>\text{Cl}^-</math>) to form a white precipitate of silver chloride (<math>\text{AgCl}</math>). When all chloride ions have reacted, excess silver ions react with potassium dichromate, producing a red-brown color that signals the endpoint.</li> </ul> |
|                                                           | <b>Oil content</b> <ul style="list-style-type: none"> <li>• Total oil content is done via immersing samples (usually metals) in a non-polar solvent for a few hours allows extraction of non-polar compounds. Non-polar solvents are used since they are effective at dissolving hydrophobic substances like oil.</li> <li>• Then transfer the solution to a clean beaker and remove any remaining metal ensures that no residual contaminants or unwanted particles affect the analysis.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

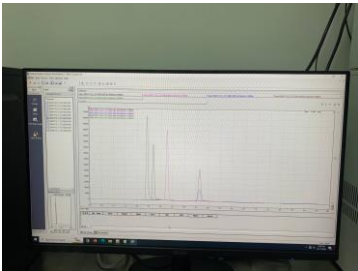
- The solution then evaporated in heated water bath at certain temperature and the remaining extracted oil is placed in an oven at a controlled temperature and allow cool to room temperature. Then record weights to determine if the samples have achieved a constant weight.

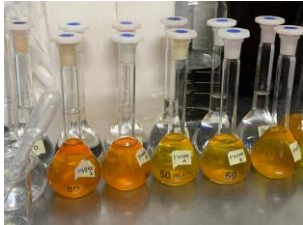




**Date/week:**  
**WEEK 9 (11-15/11/2024)**

| <b>11 November 2024</b>                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <b>BWK40103</b><br>(Quality Control and Assurance)        | <ul style="list-style-type: none"> <li>• Calibrate the pH meter before each testing session by using standardized buffer solutions (pH 4, 7, and 10 buffers) to ensure accuracy.</li> <li>• Conduct replicate tests on samples to check for repeatability and consistency of results</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>BWK40204</b><br>(Industrial Safety)                    | <ul style="list-style-type: none"> <li>• Wear a lab coat when working with chemicals to protect skin and clothing from spills and splashes, and gloves to protect from direct contact with hazardous substances</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>BWK40304</b><br>(Industrial Technology and Operations) | <p><b>pH testing</b></p> <ul style="list-style-type: none"> <li>• Weigh food samples and dissolve them in distilled water.</li> <li>• Calibrate the pH meter using standard buffer solutions begin with the pH 4 buffer solution. Rinse the electrode with distilled water and immerse it in the pH 4 buffer for calibration purpose.</li> <li>• Rinse the electrode again and proceed with the calibration of pH 7 buffer solution.</li> <li>• Finally, rinse the electrode and calibrate with the pH 10 buffer solution.</li> <li>• Rinse the pH electrode with distilled water and immerse it into the prepared food sample solution.</li> <li>• Allow the reading to stabilize, then record the pH and the temperature of the solution.</li> </ul> |
|                                                           | <p><b>Fat extraction</b></p> <ul style="list-style-type: none"> <li>• The total fat for food products is done via acid hydrolysis method. The food samples are added with concentrated acid and heated to certain temperature.</li> <li>• The fat will be hydrolysed, and the extraction is performed using petroleum ether.</li> <li>• After extraction for three times, the solvent is evaporated off in heated water bath and the final weight of fat is obtained by constant weight techniques.</li> </ul>                                                                                                                                                                                                                                         |
|                                                           | <p><b>Separation of Sudan III and IV</b></p> <ul style="list-style-type: none"> <li>• Sudan III and IV shows an overlapping peak when using gradient HPLC method.</li> <li>• Sudan III: 5.607 min</li> <li>• Sudan IV: 4.857 min</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                        |

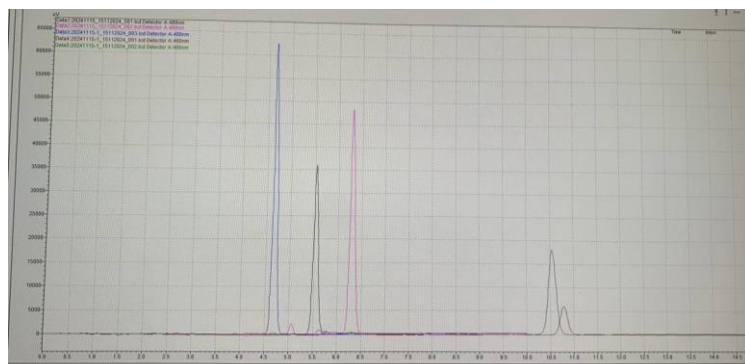
| 12 November 2024                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <b>BWK40103</b><br>(Quality Control and Assurance)        | <ul style="list-style-type: none"> <li>• Conduct replicate tests on samples to check for repeatability and consistency of results.</li> <li>• Sonicate mobile phase to degases the solvent that can interfere with the analysis. It prevents bubble formation which can cause baseline noise and peak disturbance.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>BWK40204</b><br>(Industrial Safety)                    | <ul style="list-style-type: none"> <li>• Clean all equipment thoroughly to prevent cross-contamination.</li> <li>• Label all beakers with their sample lab ID for sample identification.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>BWK40304</b><br>(Industrial Technology and Operations) | <p><b>Determine retention time of Sudan dyes</b></p> <ul style="list-style-type: none"> <li>• Sudan dyes and Para red from stock solutions diluted into 5 ppm working solutions to be injected by HPLC.</li> <li>• Gradient HPLC-UV Vis method used with mobile phase A (acetonitrile) and mobile phase B (methanol).</li> <li>• Chromatogram results show an overlapping peak between Sudan III and IV at 4.867 minute.</li> </ul>  <p>The image shows a computer monitor displaying an HPLC chromatogram. The x-axis represents time in minutes, and the y-axis represents detector response. A single, sharp, prominent peak is visible at a retention time of 4.867 minutes, indicating the co-elution of Sudan III and IV.</p> <p><b>Sample preparation for benzoic acid and sorbic acid analysis</b></p> <ul style="list-style-type: none"> <li>• The sample is first dissolved in an appropriate volume of alcohol and enhanced dissolution process via sonication, which applies ultrasonic waves to the solution.</li> <li>• After sonication, the sample is subjected to centrifugation for separating the dissolved analytes from any remaining undissolved particles, which settle at the bottom of the centrifuge tube.</li> <li>• The clear supernatant is then transferred and diluted to a specified final volume with alcohol. This "dilution to mark" ensures the sample is at a precise concentration suitable for further analysis using HPLC.</li> </ul>  <p>The image shows four Erlenmeyer flasks containing a yellow liquid, likely the prepared sample, arranged in a red plastic rack. The flasks are labeled with handwritten text, and the liquid level is visible in each.</p> |

| 13 November 2024                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| <b>BWK40103</b><br>(Quality Control and Assurance)        | <ul style="list-style-type: none"> <li>• Calibrate the pH meter before each testing session by using standardized buffer solutions (pH 4, 7, and 10 buffers) to ensure accuracy.</li> <li>• Conduct replicate tests on samples to check for repeatability and consistency of results</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>BWK40204</b><br>(Industrial Safety)                    | <ul style="list-style-type: none"> <li>• Wear a lab coat to protect against spills and chemical exposure.</li> <li>• Always wear appropriate gloves when handling chemicals to prevent skin contact.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>BWK40304</b><br>(Industrial Technology and Operations) | <b>Sample preparation for benzoic acid and sorbic acid</b> <ul style="list-style-type: none"> <li>• The sample is first dissolved in an appropriate volume of alcohol and enhanced dissolution process via sonication, which applies ultrasonic waves to the solution.</li> <li>• After sonication, the sample is subjected to centrifugation for separating the dissolved analytes from any remaining undissolved particles, which settle at the bottom of the centrifuge tube.</li> <li>• The clear supernatant is then transferred and diluted to a specified final volume with alcohol. This "dilution to mark" ensures the sample is at a precise concentration suitable for further analysis using HPLC.</li> </ul>  |
|                                                           | <b>pH testing</b> <ul style="list-style-type: none"> <li>• Weigh food samples and dissolve them in distilled water.</li> <li>• Calibrate the pH meter using standard buffer solutions begin with the pH 4 buffer solution. Rinse the electrode with distilled water and immerse it in the pH 4 buffer for calibration purpose.</li> <li>• Rinse the electrode again and proceed with the calibration of pH 7 buffer solution.</li> <li>• Finally, rinse the electrode and calibrate with the pH 10 buffer solution.</li> <li>• Rinse the pH electrode with distilled water and immerse it into the prepared food sample solution.</li> <li>• Allow the reading to stabilize, then record the pH and the temperature of the solution.</li> </ul>                                                                |
|                                                           | <b>Total oil content</b> <ul style="list-style-type: none"> <li>• Total oil content is done via immersing samples (usually metals) in a non-polar solvent for a few hours allows extraction of non-polar compounds. Non-polar solvents are used since</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

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|                                                           | <p>they are effective at dissolving hydrophobic substances like oil.</p> <ul style="list-style-type: none"> <li>• Then transfer the solution to a clean beaker and remove any remaining metal ensures that no residual contaminants or unwanted particles affect the analysis.</li> <li>• The solution then evaporated in heated water bath at certain temperature and the remaining extracted oil is placed in an oven at a controlled temperature and allow cool to room temperature. Then record weights to determine if the samples have achieved a constant weight.</li> </ul> |
| <b>14 November 2024</b>                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>BWK40103</b><br>(Quality Control and Assurance)        | <ul style="list-style-type: none"> <li>• For every batch of 10 samples, create and analyze a duplicate of one randomly selected sample within that batch to confirm the method's reliability and accuracy.</li> </ul>                                                                                                                                                                                                                                                                                                                                                               |
| <b>BWK40204</b><br>(Industrial Safety)                    | <ul style="list-style-type: none"> <li>• Wear a lab coat to protect against spills and chemical exposure.</li> <li>• Always wear appropriate gloves when handling organic acids to prevent skin contact.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                 |
| <b>BWK40304</b><br>(Industrial Technology and Operations) | <p><b>Moisture testing</b></p> <ul style="list-style-type: none"> <li>• Weigh food samples for moisture testing and dry the samples in oven for few hours to ensure that all free moisture has been fully removed, leading to a stable, accurate, and reproducible measurement.</li> <li>• Then, cooling the samples to room temperature and the final weight of moisture is obtained by constant weight techniques.</li> </ul>                                                                 |
|                                                           | <p><b>Fat extraction</b></p> <ul style="list-style-type: none"> <li>• The total fat for food products is done via acid hydrolysis method. The food samples are added with concentrated acid and heated to certain temperature.</li> <li>• The fat will be hydrolysed, and the extraction is performed using petroleum ether.</li> <li>• After extraction for three times, the solvent is evaporated off in heated water bath and the final weight of fat is obtained by constant weight techniques.</li> </ul>                                                                      |
|                                                           | <p><b>HPLC column conditioning</b></p> <ul style="list-style-type: none"> <li>• Installed new column in the HPLC system by ensuring connections were secure with no leaks.</li> <li>• Flushed column with mobile phase water: acetonitrile (95:5) for 29.05 minutes to remove any solvent and further with</li> </ul>                                                                                                                                                                                                                                                               |

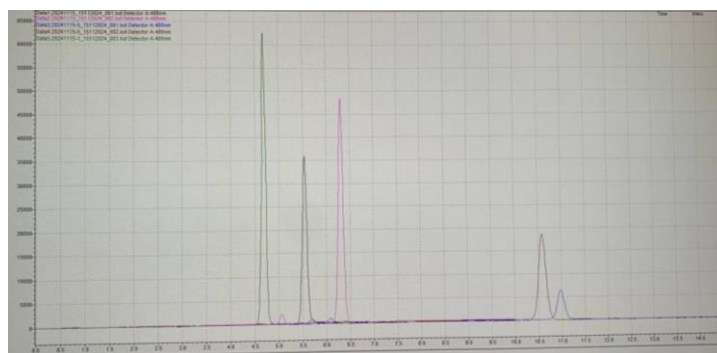
|                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                           | <p>acetonitrile 100%.</p> <ul style="list-style-type: none"> <li>It will help to ensure that the column will give a good baseline when the system is run.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>15 November 2024</b>                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>BWK40103</b><br>(Quality Control and Assurance)        | <ul style="list-style-type: none"> <li>For every batch of 10 samples, create and analyze a duplicate of one randomly selected sample within that batch to confirm the method's reliability and accuracy.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>BWK40204</b><br>(Industrial Safety)                    | <ul style="list-style-type: none"> <li>Perform the procedure in a fume hood to minimize inhalation of fumes from hydrochloric acid during the preparation of fat testing.</li> <li>Regularly check the temperature settings of the water bath to maintain it temperature, as overheating can lead to hazardous conditions or degradation of samples</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>BWK40304</b><br>(Industrial Technology and Operations) | <p><b>Determining retention time of Sudan dyes and Para red</b></p> <ul style="list-style-type: none"> <li>The retention times for each dye were determined using a gradient-HPLC method. Sudan I, Sudan II, and Para Red displayed well-separated peaks, demonstrating good resolution between these dyes. However, Sudan III and Sudan IV presented overlapping peaks, as both were eluted at the same retention time.</li> <li>To address this issue, the gradient composition of Mobile Phase B (methanol) was adjusted at the 8-minute mark to 20% methanol, aiming to improve the separation of Sudan III and IV. While this adjustment showed some improvement, the peaks for Sudan III and IV still exhibited slight overlap, indicating further optimization may be necessary for complete resolution.</li> </ul> |
|                                                           | <p><b>Fat extraction</b></p> <ul style="list-style-type: none"> <li>The total fat for food products is done via acid hydrolysis method. The food samples are added with concentrated acid and heated to certain temperature.</li> <li>The fat will be hydrolysed, and the extraction is performed using petroleum ether.</li> <li>After extraction for three times, the solvent is evaporated off in heated water bath and the final weight of fat is obtained by constant weight techniques.</li> </ul>                                                                                                                                                                                                                               |
| <b>Detail Notes of Activities</b>                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Gradient HPLC method</b><br><b>1.</b>                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| Time  | Acetonitrile | Methanol |
|-------|--------------|----------|
| 0.01  | 30           | 70       |
| 1.00  | 90           | 10       |
| 4.00  | 50           | 50       |
| 10.00 | stop         |          |



2.

| Time  | Acetonitrile | Methanol |
|-------|--------------|----------|
| 0.01  | 30           | 70       |
| 1.00  | 90           | 10       |
| 4.00  | 50           | 50       |
| 8.00  | 80           | 20       |
| 10.00 | stop         |          |



### For Industry Use

**Industry Coach's Signature:**

**Name:** Mr. Teh Yu Jie

**Date:**



**Performance and student report:**

| Criteria                             | Very Unsatisfied | Unsatisfied | Satisfied | Good | Very good |
|--------------------------------------|------------------|-------------|-----------|------|-----------|
| Meeting with supervisor              |                  |             | ✓         |      |           |
| Records of activities in the logbook |                  |             |           | ✓    |           |
| Attitude                             |                  |             |           | ✓    |           |
| Project planning                     |                  |             |           | ✓    |           |
| Management                           |                  |             | ✓         |      |           |

|                                                                                                               |
|---------------------------------------------------------------------------------------------------------------|
| <b>Comments:</b>                                                                                              |
| <b>Verified by:</b><br><br><b>Industry Coordinator's</b><br><b>Signature:</b><br><b>Name:</b><br><b>Date:</b> |